Tema 10

Integrales Dobles y Triples

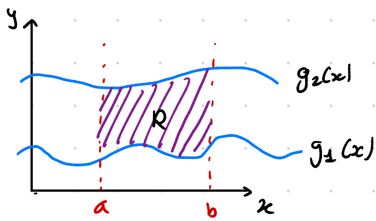
Integrales Multiples

1D → Integración
- → Area Bajo la curva
- → Obtención de función primitiva

Para Primitiva: $\int f_x(x,y)\,dx = g(x,y) + f(y)$ → Es complicado determinar $f(y)$

Area: $\iint f(x,y)\,dx\,dy$ → Integrales dobles / Iteradas → $\int_a^b \left[\int_{f_1(y)}^{f_2(y)} f(x,y)\,dx\right] dy$

Area de una Región Plana

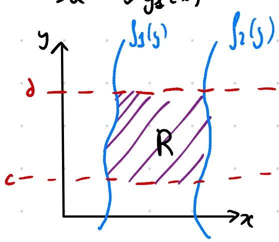


$R = [a,b] \times [g_1(x), g_2(x)]$

$\forall\ a < x < b$

$\forall\ g_1(x) < y < g_2(x)$

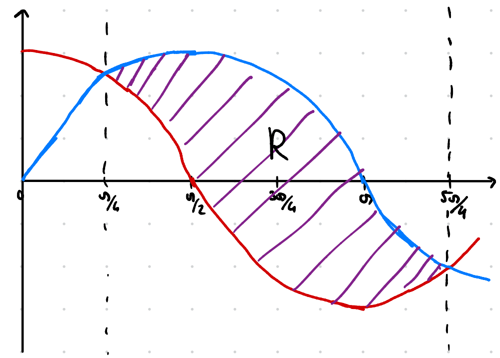
$$\int_a^b \int_{g_1(x)}^{g_2(x)} 1\,dy\,dx = \text{Area} \rightarrow \int_a^b [y]_{g_1(x)}^{g_2(x)}\,dx = \int_a^b [g_2(x) - g_1(x)]\,dx$$



$$\int_c^d \int_{f_1(y)}^{f_2(y)} dx\,dy \rightarrow \text{Area} \rightarrow \int_c^d (f_2(y) - f_1(y))\,dy$$

Esto son regiones elementales

Ej: $x \in [\pi/4, 5\pi/4]$, $y \in [\cos(x), \text{sen}(x)]$ $\rightarrow$ $R = [\pi/4, 5\pi/4] \times [\cos(x), \text{sen}(x)]$



$$\text{Area} = R = \int_{\pi/4}^{5\pi/4} \int_{\cos(x)}^{\text{sen}(x)} 1 \, dy \, dx$$

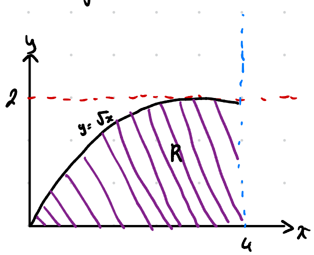
$$\text{Area} = \int_{\pi/4}^{5\pi/4} [\text{sen}(x) - \cos(x)] \, dx = \Big[-\cos(x) - \text{sen}(x) \Big]_{\pi/4}^{5\pi/4}$$

$$\text{Area} = 2\sqrt{2} \ u^2$$

El orden de la integración es arbitrario:

$$\iint f(x,y) \, dx \, dy = \iint f(x,y) \, dy \, dx$$

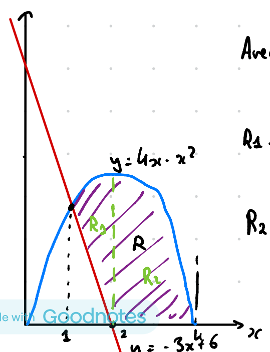
Ej: $\int_0^2 \int_{y^2}^4 dx \, dy = \text{Area}$



$$\text{Area} = \int_0^2 (4-y^2) \, dy = \Big[4y - y^3/3 \Big]_0^2 = 16/3$$

$$\text{Area} = \int_0^4 \int_0^{\sqrt{x}} dy \, dx = \int_0^4 \sqrt{x} \, dx = \Big[\tfrac{2}{3} x^{3/2} \Big]_0^4 = 16/3 \ u^2$$

Ej: Area entre $y = 4x - x^2$ e $y = -3x + 6$. Hay un cambio en la función inferior



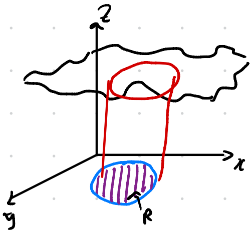
$$\text{Area} = R_1 + R_2 = \int_1^2 \int_{-3x-6}^{4x-x^2} dy \, dx + \int_2^4 \int_0^{4x-x^2} dy \, dx = 15/2$$

$$R_1 = \int_1^2 \int_{-3x-6}^{4x-x^2} dy \, dx$$

$$R_2 = \int_2^4 \int_0^{4x-x^2} dy \, dx$$

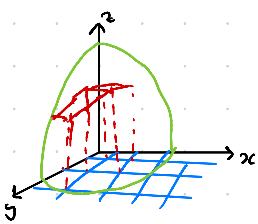
Integrales dobles y Volumen

Dada una función continua $f(x,y) > 0$ $\forall (x,y)$ en una región del xy



$$\iint_R f(x,y)\, dy\, dx \to \text{Volumen}$$

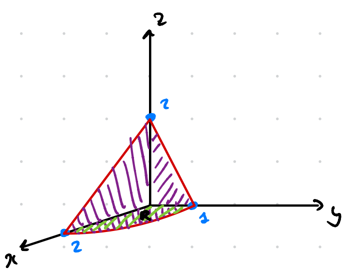
Queremos calcular el volumen debajo de la superficie $z = f(x,y)$ y el plano xy.



$$\text{Suma de Reimann} : \sum_{i=1}^{N} f(x_i, y_i)\, \Delta V_i$$

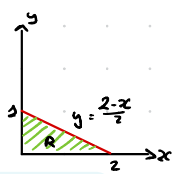
$$\text{Volumen} = \lim_{|V_i| \to 0} \sum_{i=1}^{N} f(x_i, y_i)\, \Delta V_i$$

Ej: Volumen Entre el Plano $z = 2 - x - 2y$ y los tres planos de coordenadas



$$z = 0 \to y = \frac{-x+2}{2} \begin{cases} x = 0 \to y = 1 \\ y = 0 \to x = 2 \end{cases}$$

$$x = y = 0 \to z = 2$$



$$R = \int_0^2 \int_0^{\frac{2-x}{2}} 1\, dy\, dx$$

$$\text{Volumen} = \int_0^2 \int_0^{\frac{2-x}{2}} (2 - x - 2y)\, dy\, dx = 2/3\ u^3$$

Teorema de Fubini

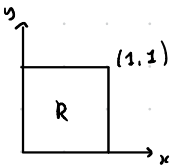
a) Si $R = [a,b] \times [g_1(x), g_2(x)]$ $\quad g_1$ y g_2 continuas en $[a,b]$

$$\iint_R f(x,y)\,dy\,dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,\underbrace{dy\,dx}_{dA}$$

b) si $R = [h_1(y), h_2(y)] \times [c,d]$

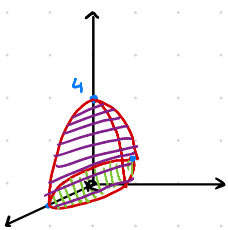
$$\iint_R f(x,y)\,dx\,dy = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$

Ej: $\iint \left(1 - \frac{1}{2}x^2 - \frac{1}{2}y^2\right) dA \qquad R = [0,1] \times [0,1]$



$$\int_0^1 \int_0^1 \left(1 - \frac{x^2}{2} - \frac{y^2}{2}\right) dA = \frac{2}{3}$$

Ej: Volumen de Paraboloide $z = 4 - x^2 - 2y^2$ y el plano xy



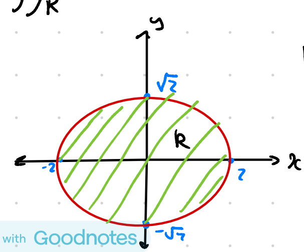
$x = y = 0 \to z = 4$

$z = 0 \to x^2 - 2y^2 = -4 \to y^2 = \frac{x^2+4}{2} \to y = \pm\sqrt{\frac{x^2+4}{2}}$

$x = 0 \to y = \pm\sqrt{2}$

$y = 0 \to x = \pm 2$

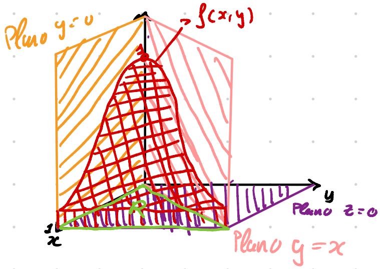
$$\iint_R f(x,y)\,dx\,dy$$



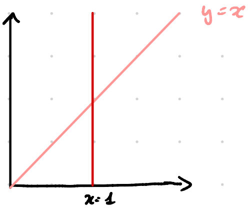
$$V = \int_{-2}^{2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{\sqrt{\frac{4-x^2}{2}}} (4 - x^2 - 2y^2) = \frac{4}{3\sqrt{2}} \int_{-2}^{2} (4 - x^2)^{3/2}\,dx$$

$$V = 4\sqrt{2}\,\pi$$

Ej: Volumen contenido entre $f(x,y) = e^{-x^2}$ y los planos $z=0$, $y=0$, $y=x$ y $x=1$



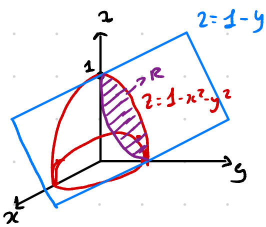
$$\iint_R e^{-x^2}\,dy\,dx$$



$$\int_0^1 \int_0^x e^{-x^2}\,dy\,dx = \frac{e-1}{2e}$$

$$\int_0^1 \int_y^1 e^{-x^2}\,dx\,dy \Rightarrow \nexists \int e^{-x^2}dx$$

Ej: Volumen por debajo del paraboloide $z = 1 - x^2 - y^2$ y por encima de plano $z = 1-y$

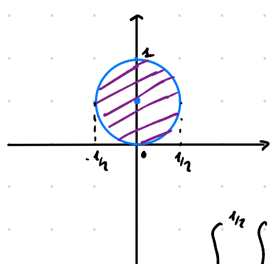


$x=0=y \rightarrow z=1 \qquad z=0 \rightarrow x^2+y^2=1$

Intersección paraboloide con el plano: $1-x^2-y^2 = 1-y \rightarrow x^2 = y-y^2 \Rightarrow x^2 = -1(y^2-y)$

$x^2 = -(y-\frac{1}{2})^2 + \frac{1}{4} \rightarrow x^2 + (y-\frac{1}{2})^2 = \frac{1}{4}$

Círculo de radio 0,5 con centro en $(0, \frac{1}{2})$

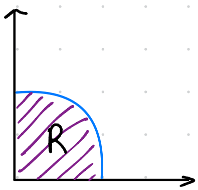


$$\int_{-1/2}^{1/2} \int_{-\sqrt{y-y^2}}^{\sqrt{y-y^2}} 1-x^2-y^2\,dy\,dx = \frac{51}{32}$$

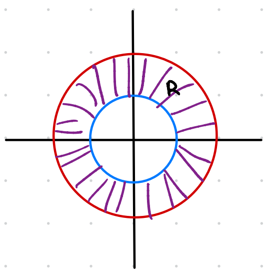
Cambio de Coordenadas

Coordenadas Polares:

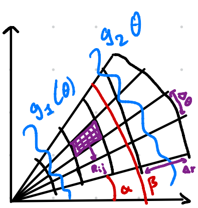
- $x = r\cos\theta$, $y = r\,sen(\theta)$, $r = \sqrt{x^2+y^2}$, $tg(\theta) = y/x$



$R = \{(r,\theta) : 0 \le r \le 2 ;\ 0 \le \theta \le \pi/2)$



$R = \{(r,\theta) : 1 \le r \le 2 ;\ 0 \le \theta \le 2\pi \}$

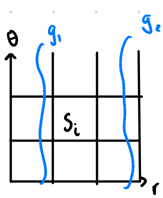


Región: $r_1 = g_1(\theta)$ $\quad \theta = \alpha$

$r_2 = g_2(\theta)$ $\quad \theta = \beta$

Area $R_{ij} = r_i \Delta r_i \Delta\theta_i$ $\quad \Delta r = r_2 - r_1$ $\quad \Delta\theta = \theta_2 - \theta_1$

¿$dx\,dy = dr\,d\theta$? $\quad$ Volumen $\simeq \sum f(r_i\cos\theta_i , r_i\,sen\,\theta_i)\, r_i \Delta r_i \Delta\theta_i$



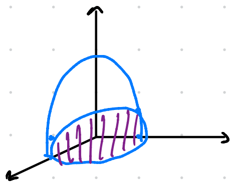
= Suma de Riemann

Sector polar se corresponde a los rectangulos si $\Delta A_i = \Delta r_i \Delta\theta_i$

$$\sum f(r_i\cos\theta_i , r_i\,sen(\theta_i))\, r_i \Delta r_i \Delta\theta_i = \iint f(r_i\cos(\theta_i), r_i\,sen(\theta_i))\, r\, dA$$

Entonces: $\iint_R f(x,y)\,dx\,dy = \int_{\alpha}^{\beta}\int_{g_1(\theta)}^{g_2\theta} f(r\cos(\theta), r\,\mathrm{sen}(\theta))\, r\,dr\,d\theta$

Ej: Volumen entre el plano $z=0$ y $z = 1-x^2-y^2$

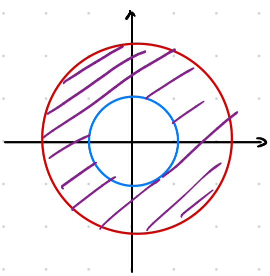


$R = \{(r,\theta) : 0 \le r \le 1 ;\ 0 \le \theta \le 2\pi\}$

$V = \int_0^1 \int_0^{2\pi} (1-r^2) r\,d\theta\,dr = \int_0^{2\pi} d\theta \int_0^1 (1-r^2) r\,dr$

$V = \pi/2$

Ej: Region Angular entre los circulos $x^2+y^2=1$ y $x^2+y^2=5$

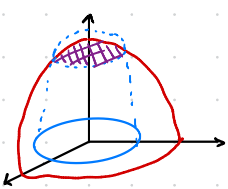


$R = \{(r,\theta) : 1 \le r \le \sqrt{5} ;\ 0 \le \theta \le 2\pi\}$

$\int_0^{2\pi}\int_1^{\sqrt{5}} (r^2\cos^2\theta + r\,\mathrm{sen}\,\theta)\, r\,dr\,d\theta = 6\pi$

$\iint_R (x^2+y)\,dA$

Ej: Volumen entre el hemisferio $z = (16-x^2-y^2)^{1/2}$ y la region circular $x^2+y^2=4$

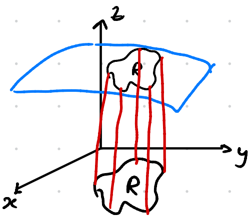


$R = \{[0,2]\times[0,2\pi]\}$

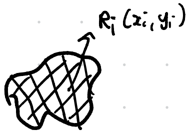
$R = \{(r,\theta) : 0 \le r \le 2 ;\ 0 \le \theta \le 2\pi\}$

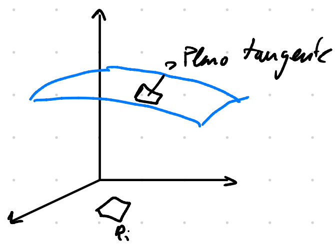
$V = \int_0^{2\pi}\int_0^2 (16 \underbrace{- r^2\cos^2\theta - r^2\mathrm{sen}^2\theta}_{-r^2})\, r\,dr\,d\theta = \int_0^{2\pi}\int_0^2 (16-r^2)\,dr\,d\theta = {}^{16\pi}/_{3}\,(8-3\sqrt{3})$

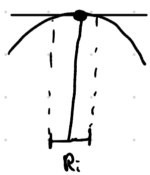
Área de una Superficie



1. Particionamos R en n rectangulos







Construimos el plano tangente que pasa por el plano $(x_i, y_i, f(x_i, y_i))$

Area $R_i \Rightarrow \Delta A_i = \Delta x_i \Delta y_i$ $\quad \Delta A_i \approx \Delta T_i$ $\quad$ Area Total $\Rightarrow \sum_{i=1}^{n} \Delta A_i \cong \sum_{i=1}^{n} \Delta T_i$

2. Calcular el plano tangente

3. Calcular el area de $\Delta T_i \rightarrow$ Calcular sus lados

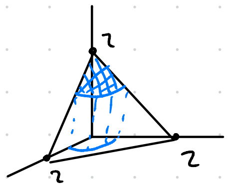
$$\vec{u} = \Delta x_i \vec{i} + f_x(x_i, y_i)\Delta y_i \vec{k}$$
$$\vec{v} = \Delta y_i \vec{j} + f_y(x_i, y_i)\Delta y_i \vec{k}$$

Calcular el producto vectorial: $\vec{u} \times \vec{v} = \vec{n} = \langle -f_x, -f_y, 1 \rangle \Delta x_i \Delta y_i$

$$\|\Delta T_i\| = \|\vec{u} \times \vec{v}\| = (f_x^2 + f_y^2 + 1)^{1/2} \, \Delta A_i$$

Superficie $= \iint_R (f_x^2+f_y^2+1)^{1/2}\, dA$

Ej: Calcular la superficie del plano $z = 2-x-y$ por encima de $x^2+y^2 \le 1$

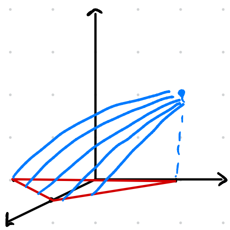


$$S = \iint_R (f_x^2 + f_y^2 + 1)^{1/2}\, dx\, dy$$

$$S = \iint_R 3^{1/2}\, dx\, dy$$

$$R = \{(r,\theta) : 0 < r < 1,\ 0 \le \theta \le \pi/2\} \Rightarrow S = \sqrt{3}\,\pi/2 \left(\tfrac{1}{2} r^2 \Big|_0^1\right) = \sqrt{3}\,\pi/4$$

Ej: $f(x,y) = 1 - x^2 - y$ $R =$ triangulo con vertices $(1,0,0)$, $(0,-1,0)$, $(0,1,0)$



$f_x = -2x$

$f_y = -1$

$$S = \iint_R \left(4x^2 + 1 + 1\right)^{1/2}$$

$$= \iint_R (4x^2+2)^{1/2}\, dx\, dy$$

$$= \int_0^1 \int_{x-1}^{1-x} (4x^2+2)^{1/2}$$

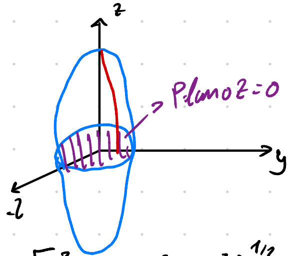
Integrales Triples

$$\iiint_Q f(x,y,z)\, dx\, dy\, dz = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x,y)}^{g_2(x,y)} f(x,y,z)\, dz\, dy\, dx$$

$f(x,y,z) = w$

$$Q = \begin{cases} a \le x \le b \\ h_1(x) \le y \le h_2(x) \\ g_1(x,y) \le z \le g_2(x,y) \end{cases}$$

Ej: Calcular el volumen del elipsoide $4x^2+4y^2+z^2=16$



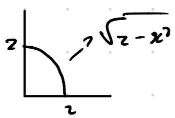
Nos limitamos al primer cuadrante

$V=\iiint_Q dx\,dy\,dz$ -> Estado cuando $z=0$

$$8\times\int_0^2\int_0^{\sqrt{2-x^2}}\int_0^{(16-4x^2-4y^2)^{1/2}} dz\,dx\,dy$$

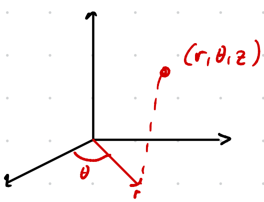
$z=0$ en la función $f(x,y,z)$ obtenemos la proyección en el plano xy.

$R \to 4x^2+4y^2=16 \to x^2+y^2=4 \to$ circulo de radio 2



$V=\frac{64\pi}{3}$

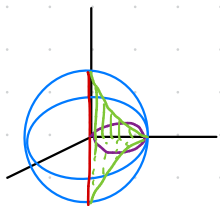
Integrales Triples con coordenadas Cilíndricas



$$\iiint_Q f(x,y,z) = \int_{\theta_1}^{\theta_2}\int_{g_1}^{g_2(\theta)}\int_{h_1(r\cos\theta,\, r\,\text{sen}\,\theta)}^{h_2(r\cos\theta,\, r\,\text{sen}\,\theta)} f(r\cos\theta,\, r\,\text{sen}(\theta),\, z)$$

Ej: Dada la esfera $x^2+y^2+z^2=4$ y el cilindro $r=2\,\text{sen}\,\theta$.

Calcular V



Intersección entre esfera y el cilindro

$$x^2+y^2+z^2=4 \rightarrow r^2+z^2=4$$

$$V=\int_0^{\pi}\int_0^{2\,\text{sen}\,\theta}\int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} r\,dz\,dr\,d\theta$$

$$V=\frac{16}{9}(3\pi-4)$$

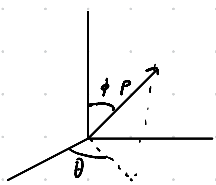
Integrales triples en coordenadas esféricas

$$\iiint_Q f(x,y,z)\,dx\,dy\,dz=\int_{\theta_1}^{\theta_2}\int_{\phi_1}^{\phi_2}\int_{\rho_1}^{\rho_2} f(\rho\,\text{sen}\,\phi\cos\theta,\ \rho\,\text{sen}\,\phi\,\text{sen}\,\theta,\ \rho\cos\theta)\ \rho^2\,\text{sen}\,\phi\ d\rho\,d\phi\,d\theta$$

$x=\rho\,\text{sen}\,\phi\cos\theta$

$y\ \rho\,\text{sen}\,\phi\,\text{sen}\,\theta$

$z=\rho\cos\phi$



Cambio de Variable

Función de 1 variable $\rightarrow \int_a^b f(x)dx = \begin{cases} x = g(u) \\ dx = g'(u)du \end{cases}$

$$\int_{g(a)}^{g(b)} f(g(u))\, g'(u)\, du$$

Funciones de varias variables:

cartesianas →
- Polares (2D) → $dx\,dy \rightarrow r\,dr\,d\theta$
- Cilindricas (3D) → $dx\,dy\,dz \rightarrow r\,dr\,d\theta\,dz$
- esfericas (3D) → $dx\,dy\,dz \rightarrow \rho^2 \operatorname{sen}\phi\, d\theta\, d\phi\, d\rho$

Jacobiano

Función de 2 variables, $f(x,y)$

$$\left.\begin{aligned} x &= g(u,v) \\ y &= h(u,v) \end{aligned}\right\} \quad \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\cdot\frac{\partial y}{\partial v} - \frac{\partial y}{\partial u}\cdot\frac{\partial x}{\partial v}$$

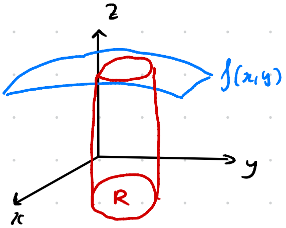
Cartesianas a Polares

$$\left.\begin{aligned} x &= r\cos\theta \\ y &= r\operatorname{sen}\theta \end{aligned}\right\} \quad \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\operatorname{sen}\theta \\ \operatorname{sen}\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\operatorname{sen}^2\theta = r$$

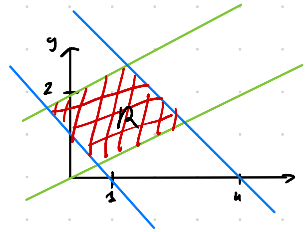
Caso General:

$$\iint f(x,y)\,dx\,dy \xrightarrow[]{\substack{x = g(u,v) \\ y = h(u,v)}} \iint f(g(u,v), h(u,v)) \frac{\partial(x,y)}{\partial(u,v)}\,du\,dv$$

Ej: $\iint_R f(x,y)\,dx\,dy$



$$R = \begin{cases} x - 2y = 0 & x - 2y = -4 \\ x + y = 4 & x + y = 1 \end{cases}$$



Cambio de variable:

$u = x + y \qquad x = \frac{1}{3}(2u + v)$

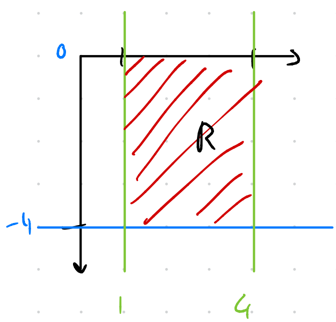
$v = x - 2y \qquad y = \frac{1}{3}(u - v)$

$x - 2y = 0 \rightarrow v = 0$

$x - 2y = 4 \rightarrow v = -4$

$x + y = 1 \rightarrow u = 1$

$x + y = 4 \rightarrow u = 4$

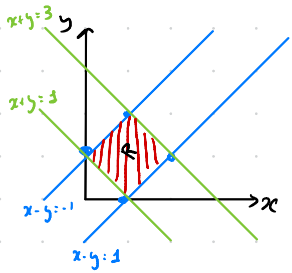


Jacobiano $\Rightarrow \dfrac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{vmatrix} = -\frac{1}{3} \rightarrow \iint_R 3xy\,dy \rightarrow$

$$\int_1^4 \int_{-4}^0 3 \cdot \frac{1}{3}(2u+v) \cdot \frac{1}{3}(u-v) \cdot \left(-\frac{1}{3}\right) du\,dv = \int_1^4\int_{-4}^0 -\frac{1}{9}(2u+v)(u-v)\,du\,dv = \frac{164}{9}$$

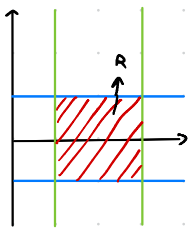
Ej: $\iint_R (x+y)^2 \operatorname{sen}^2(x-y)\,dx\,dy$

R = Región contenida entre los vertices $(0,1), (1,2), (2,1), (1,0)$



$u = x+y \qquad 1 \leq u \leq 3$

$v = x-y \qquad -1 \leq v \leq$



$x = \frac{1}{2}(u+v)$

$y = \frac{1}{2}(u-v)$

Jacobiano $\rightarrow \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{2}$

$$\int_{-1}^{1}\int_{1}^{3} u^2 \operatorname{sen}^2(v)\left(-\frac{1}{2}\right)du\,dv = \frac{13}{6}(2-\operatorname{sen} 2)$$